# PHY101: Introduction to Physics-1

***Monsoon, 2017***

## Tutorial-9

1. The equation of motion of a particle is given by $x = 10\sin(10t + \frac{\pi}{3})$, where $x$ is in centimeter and $t$ in second. When does the particle (a) first come to rest (b) first have zero acceleration and (c) first have maximum speed?

2. The initial position and initial velocity of an object moving in simple harmonic motion are $x_i$ and $v_i$; the angular frequency of oscillation is $w$.
(a) Show that the position and velocity of the object for all time can be written as,

$$x(t) = x_i cos wt + \frac{v_i}{w} sin wt$$

$$v(t) = -x_i w sin wt + v_i cos wt$$

(b) If the amplitude and acceleration of the motion are $A$ and $a$ respectively, show that

$$v^2 - ax = v_i^2 - a_i x_i = w^2 A^2$$

3. For what value of $b$ does $\ddot{x} + b\dot{x} + x = 0$ exhibit critical damping? For this value of $b$, what is the solution $x_1$ with $x(0) = 1$ and $\dot{x}(0) = 0$? What is the solution $x_2$ with $x(0) = 0$ and $\dot{x}(0) = 1$?

4. A damped harmonic oscillator consists of a block ($m$ = 2 kg), a spring ($k$ = 30 N/m), and a damping force ($F = -bv$). Initially, it oscillates with amplitude of 25 cm; because of the damping, the amplitude falls to three-fourths of this initial value at the completion of four oscillations.
(a) What is the value of $b$? (b) How much energy has been "lost" during these four oscillations?